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(54) **PROCESS FOR THE PREPARATION OF MOLNUPIRAVIR**

VERFAHREN ZUR HERSTELLUNG VON MOLNUPIRAVIR

PROCÉDÉ DE PRÉPARATION DE MOLNUPIRAVIR

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(56) References cited:

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- **GOPALSAMUTHIRAM VIJAYAGOPAL ET AL: "A Concise Route to MK-4482 (EIDD-2801) from Cytidine: Part 2", SYNLETT, vol. 32, no. 03, 29 September 2020 (2020-09-29), DE, pages 326 - 328, XP055868434, ISSN: 0936-5214, Retrieved from the Internet <URL:https://www.thieme-connect.de/products/ejournals/pdf/10.1055/a-1275-2848.pdf> DOI: 10.1055/a-1275-2848**

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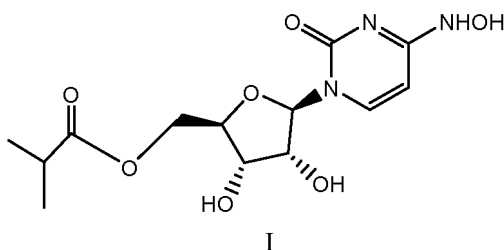
Description**Field of Invention**

[0001] The present invention relates to an improved process for the preparation of Molnupiravir (MK-4482, earlier known as EIDD-2801), an orally active antiviral agent which is being investigated for treating COVID-19 infection.

Background of the Invention

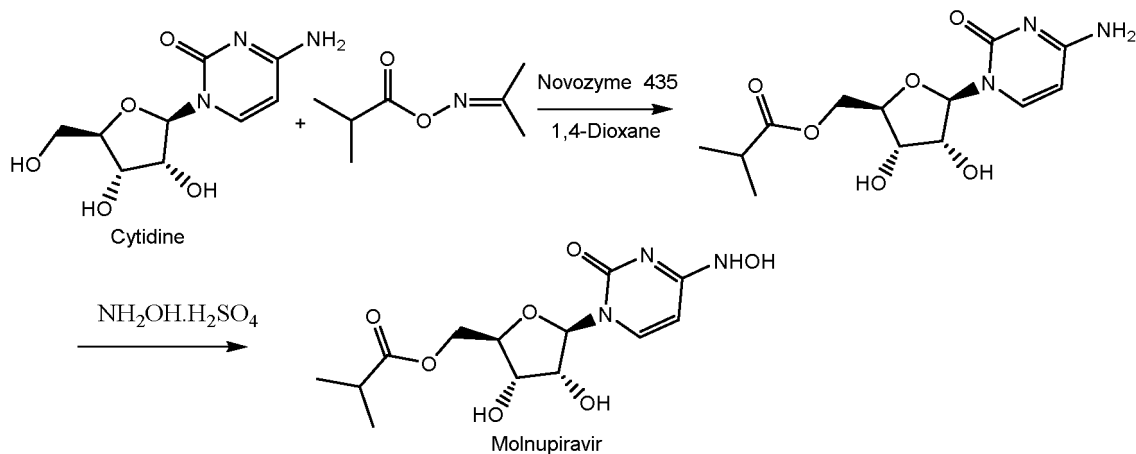
Molnupiravir is chemically [(2R,3S,4R,5R)-3,4-dihydroxy-5-[4-(hydroxyamino)-2-oxypyrimidin-1-yl]oxolan-2-yl]methyl 2-methylpropanoate, having the structure (I):

[0002]



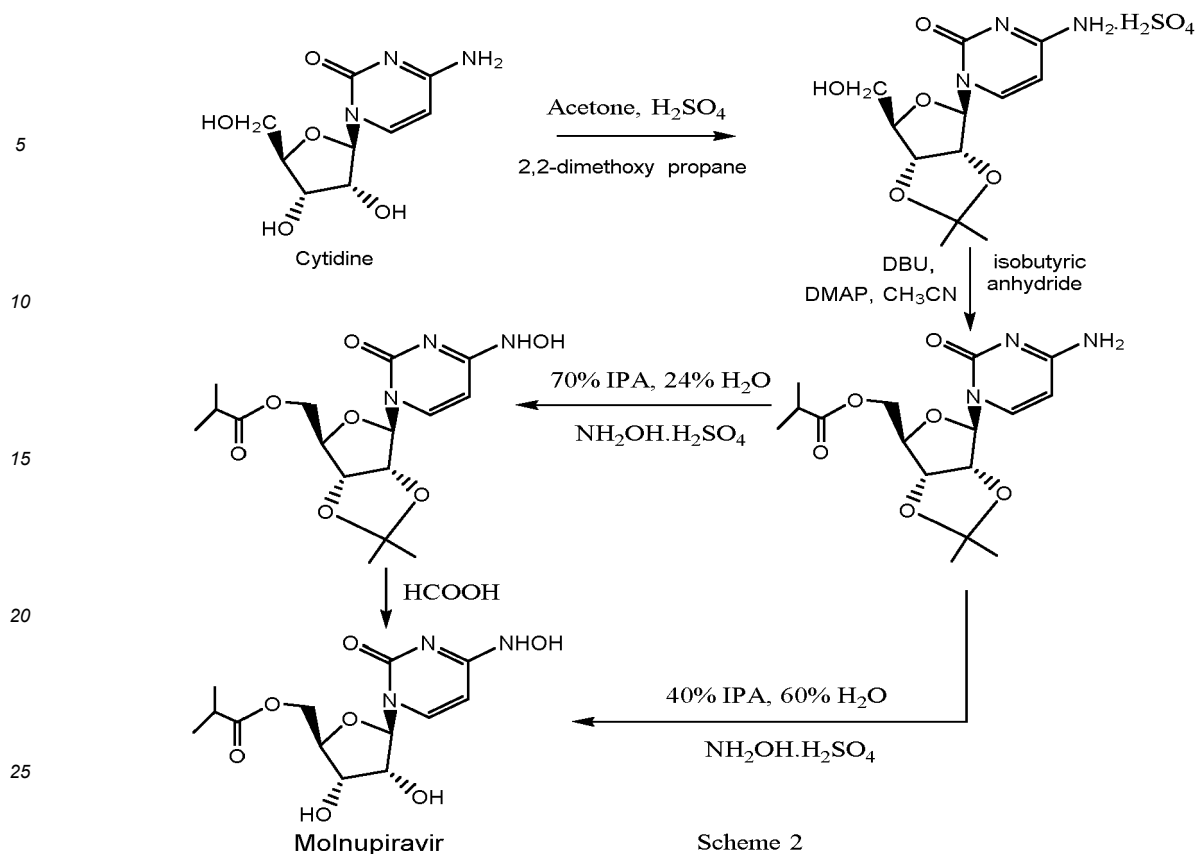
[0003] Molnupiravir was discovered at Emory University and is undergoing clinical development in partnership with Ridgeback Biotherapeutics and Merck & Co.

[0004] The first synthetic route reported for Molnupiravir used uridine as the starting material which is expensive and of limited availability [WO2019113462; WO2019173602]. Furthermore, the route suffered from low yields. Compared to uridine, cytidine is less expensive and direct transamination of cytidine with hydroxylamine is possible. Hence new routes based on cytidine have been reported (Schemes 1 & 2).



[0005] Scheme 1 discloses selective esterification of the primary alcohol of cytidine with isobutyryl acetone oxime ester using Novozyme 435 which is immobilized *Candida antarctica* Lipase B. In the next step, the cytidine ester was reacted with hydroxylamine sulphate to obtain molnupiravir (Chem. Commun. 2020, 56, 13363-13364). The use of immobilized enzyme makes the scheme expensive. Furthermore, some amounts of di and tri ester impurities are formed, in spite of using the enzyme, during the first stage and partial deamination was observed in the second stage. Both stages require purification through column chromatography.

[0006] Scheme 2 adopted chemical approach to selectively esterify primary alcohol of cytidine, instead of the enzyme system.



[0007] Cytidine was reacted with acetone, sulfuric acid and 2,2-dimethoxy propane to obtain 2',3'-O-isopropylidene cytidine sulfate. It was reacted with isobutyric anhydride in the presence of 1.8-diazabicyclo[5.4.0]undec-7-ene (DBU) and catalytic amount of 4-dimethylaminopyridine (DMAP) in acetonitrile. The resulting cytidine acetonide ester was reacted with hydroxylamine sulfate in 70% IPA (24% water by KF titration), followed by deprotection of acetonide using formic acid to give Molnupiravir. The final product was isolated using column purification in 64% yield. Alternatively, the cytidine acetonide ester was directly converted into molnupiravir by carrying out both hydroxylation and acetonide deprotection in one step by treating with hydroxylamine sulfate in 40% IPA (60 % water) for a longer time. However, this resulted in significant hydrolysis of the ester leading to the formation of about 20% N-hydroxycytidine as a byproduct (Synlett.2021, 32(3), 326-328).

[0008] Another major drawback of the Scheme 2 is that we observed release of carbon monoxide during the reaction with formic acid used in acetonide deprotection. Use of formic acid for deprotection of acetonide is reported in several literature (WO2019113462; WO2019173602; WO 2016/106050A1). However, it is also known in the literature that formic acid is a source of carbon monoxide and mere storing it for a longer time results in the liberation of carbon monoxide (containers are to be vented frequently).

[0009] Thus, both the schemes based on cytidine have certain drawbacks and there is a need for a safer alternative process.

Summary of the Invention

[0010] While studying the esterification of 2',3'-O-isopropylidene cytidine sulfate (II) with butyric anhydride in the presence of base, it was observed that, in addition to 5'-isobutyryl cytidine acetonide (III), about 8% amide is forming as a byproduct (Impurity-1) due to the cross-reaction of the amino group of the cytidine, (Scheme 3).

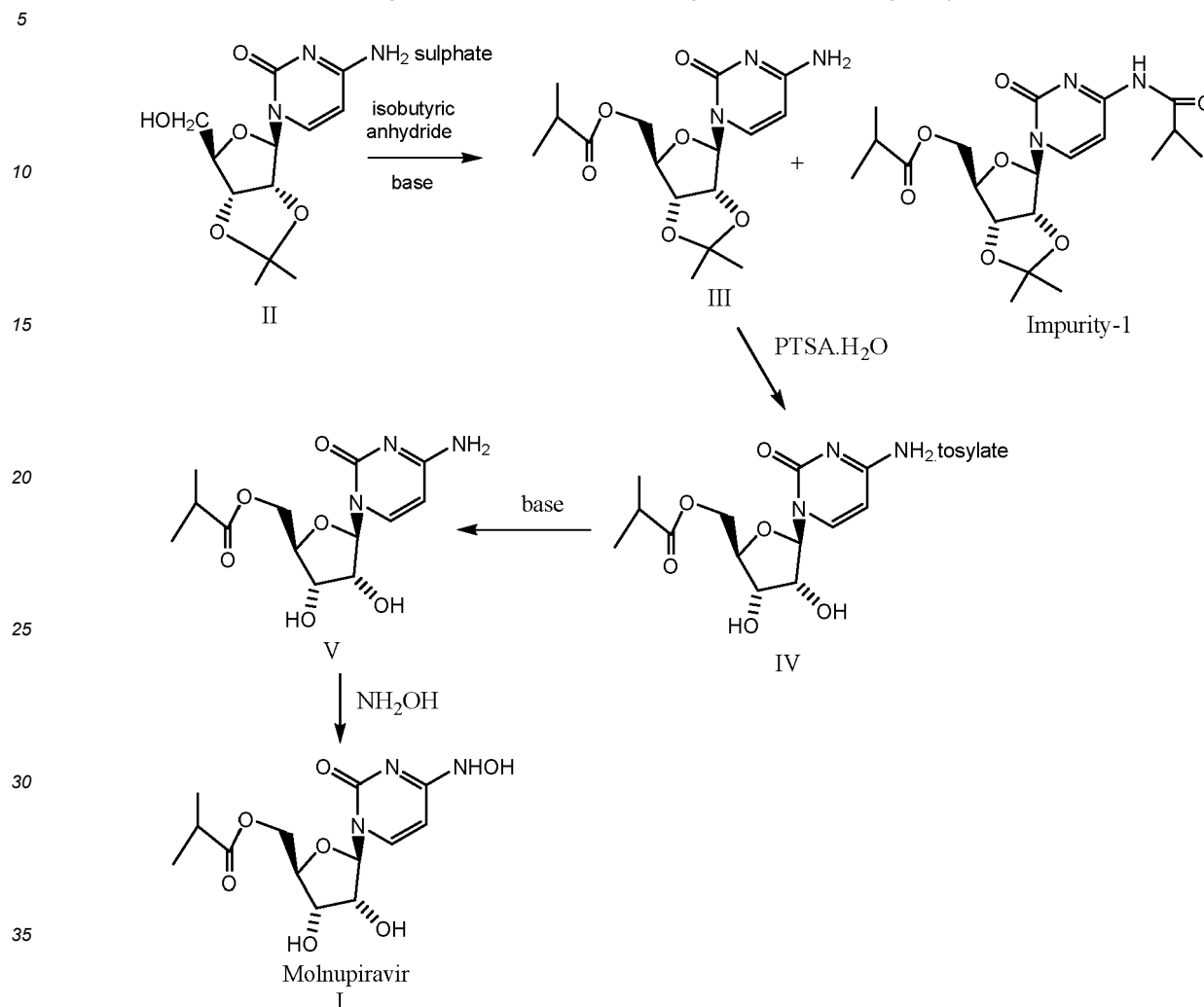
[0011] As a method of purification, without using column chromatography, salt formation with various acids were investigated. When para-toluenesulphonic acid (PTSA) which is a monohydrate was used for the salt formation, besides obtaining pure salt which was free from the amide impurity, deprotection of the acetonide group was also observed and directly 5'-isobutyrylcytidine tosylate ((IV) was obtained (Scheme 3).

[0012] The importance of the water of crystallization is further demonstrated by the fact that azeotroped PTSA was unable to hydrolyze the acetonide group, *albeit* it could form a tosylate salt.

[0013] It was also a pleasant surprise to note that the water of crystallization present in the acid was hydrolyzing only the acetonide group without affecting the isobutyryl ester group present in the molecule.

[0014] The tosylate salt (IV) was converted to 5-isobutyl cytidine free base (V), and on reaction with hydroxylamine gives Molnupiravir (I).

[0015] The main advantage of the present process is avoiding formic acid for acetonide deprotection which is a source of carbon monoxide and achieving the purification without using column chromatography.

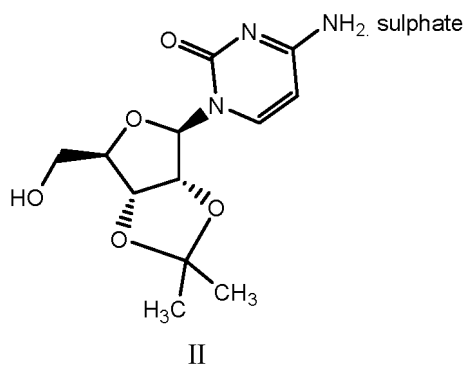


Scheme 3

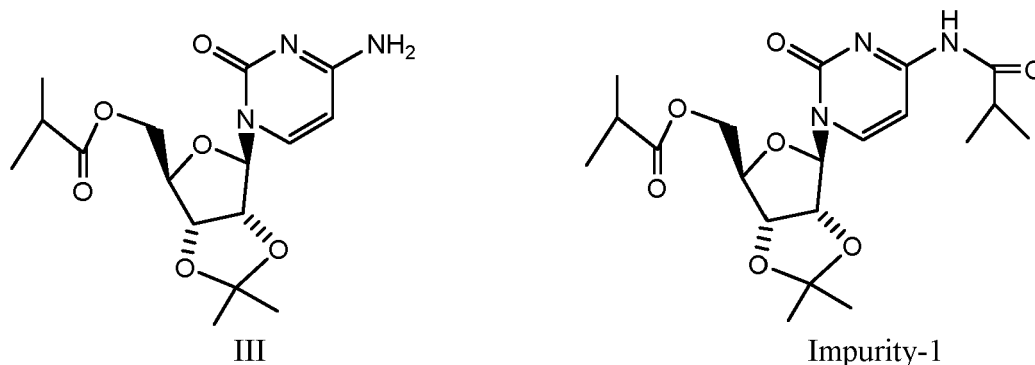
Detailed Description of the Invention

[0016] The present invention provides a novel process for the preparation of Molnupiravir which comprises the steps of:

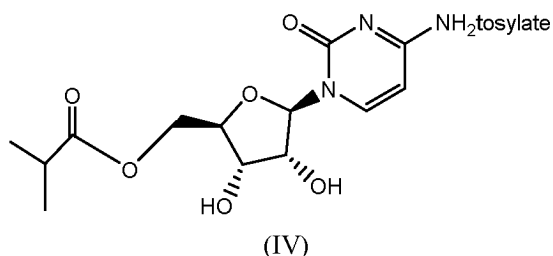
(a) reacting 2',3'-O-isopropylidene cytidine sulphate having the structure II,



with isobutyric anhydride in a solvent in the presence of a non-nucleophilic base at room temperature to obtain 5'-isobutyryl ester of the structure (III), having N-isobutyric amide as an impurity (Impurity 1),



(b) reacting the reaction product obtained from step-a with para-toluenesulphonic acid monohydrate to obtain 5'-isobutyryl cytidine acid tosylate salt (IV), which is free from N-isobutyric amide impurity, and



(c) converting 5'-isobutyryl cytidine acid salt (IV) to its free base (V) and reacting the free base with hydroxylamine to obtain Molnupiravir (I).

[0017] The required starting material, 2',3'-O-isopropylidene cytidine sulphate of formula (II), can be prepared by the method as described in Synlett (2021), 32(3), 326-328.

[0018] Esterification of (II) with isobutyric anhydride in the presence of a non-nucleophilic base gives the isobutyric ester (III). The reaction can be carried out in acetonitrile at room temperature. The non-nucleophilic bases that are used are selected from the group consisting of triethylamine, diisopropylethyl amine (DIPEA), 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), N-methyl morpholine (NMM), and 1,4-diazabicyclo-[2.2.2]octane (DABCO). Addition of catalytic amount of 4-dimethylaminopyridine enhances the reaction. The reaction takes about 18-24 hours for the completion after which it is concentrated and the residue is stirred with a mixture of water and dichloromethane for about 5 minutes. After separating the layers, the organic layer is dried and concentrated to get a foamy solid which contains about 90% yield of (III). The residue also contains Impurity-1, in about 8%, which is an amide derivative and an unknown Impurity-2 in about 0.5% in HPLC analysis.

[0019] Treating the above residue with para-toluenesulphonic acid monohydrate, resulted in obtaining 5'-isobutyryl cytidine tosylate along with the acetonide hydrolysis. The reaction can be conducted in ethyl acetate or acetonitrile. At room temperature, only tosylate salt is formed. However, when the reaction is conducted at 75-80°C for about 24 hours, the acetonide moiety gets hydrolysed to form (IV) having 98% purity by HPLC. It was completely free from Impurity-1 and the Impurity-2 was reduced to about <0.05%. The hydrolysis of the acetonide was caused by the water of crystallization present in para-toluenesulphonoc acid monohydrate which is about 12%. The PTSA after azeotrope in xylene was ineffective to cause the hydrolysis of acetonide. Drying PTSA monohydrate in desiccator did not decrease the water content. When acetonitrile was used as solvent, about 5% hydrolysis of ester was observed. Reacting the tosylate salt (IV) with a base gives 5'-isobutyryl cytidine (V). Although various organic bases can be used, best results was obtained when dicyclohexylamine (DCHA) was used. Treating IV with DCHA, about 1 to 1.5 equivalent, in acetone at room temperature for about an hour, results in the precipitation of PTSA.DCHA salt which is removed by filtration. The filtrate is concentrated and the residue is treated with water and dichloromethane, to remove the residual PTSA-DCHA salt and remaining free DCHA.

[0020] Reaction of cytosine derivatives with hydroxylamine to obtain N-hydroxy cytosines is known since 1965 (Biochemical & Biophysical research Communications, 18(4), 1965,617-622).

[0021] The free base of cytidine butyrate(V) can be converted into molnupiravir by reacting with hydroxylamine. The reaction can be conducted using commercially available salts of hydroxylamine such as either HCl or sulphate.

Completion of the reaction requires 3 to 5 moles of hydroxylamine and heating at 70 to 80 °C for 15 to 20 hours.

[0022] The embodiments of the present invention are further described in the following examples, which are not intended in any way to limit the scope of the invention.

EXAMPLES

Example-1: Preparation of 5-isobutrylcytidine tosylate (IV):

[0023] Cytidine acetonide sulphate (II) (22 g, 57.69 mmol), 4-dimethylaminopyridine (1.41g, 0.2 equiv.), 1,4-diazabicyclo[5.4.0]undec-7-ene (27.23g, 3.1 equiv.), were dissolved in acetonitrile (220 mL) and isobutyric anhydride (14.6g, 1.6 equiv.) was added during 15 min. The solution was stirred for 20h. The reaction mixture was concentrated and the residue was treated with water (100 mL) and dichloromethane (100 mL). After stirring for 5 minutes, the two layers were separated and the organic layer was dried over anhydrous sodium sulphate and concentrated to get 20.9 g foamy solid containing 89.7% III, 8.54% Impurity-1 and 0.54% an unknown impurity-2.

[0024] The above solid was dissolved in ethyl acetate (100 mL), to this was added a solution of para-toluenesulphonic acid monohydrate (10.98 g, 1.0 equiv.) in ethyl acetate (100 mL). The reaction mixture was heated to 75-80°C and stirred for 24h. After cooling to room temperature, the precipitated solids were filtered and washed with ethyl acetate (100 mL), dried under vacuum for 4 hours to obtain 21.5 g (76.7%) IV. Purity by HPLC: 98.53%, tosylate salt of III: 0.24 %, Impurity-1: nil, Impurity-2: 0.03%

[0025] FT-IR (KBr, cm^{-1}): 3481, 3415, 3282, 3137, 2923, 1724, 1690, 1542, 1497, 1453, 1420, 1397, 1331, 1280, 1243, 1198, 1172, 1138, 1126, 1099, 1035, 1010, 919, 875, 830, 814, 765, 683, 622, 585, 567, 527. $^1\text{H-NMR}$ (300 MHz, $\text{DMSO-}d_6$): δ 9.50 (s, 1H), 8.44 (s, 1H), 7.95-7.93 (d, 1H), 7.50-7.46 (d, 2H, Ar-H), 7.13-7.10 (d, 2H, Ar-H), 6.12-6.09 (d, 1H), 5.71-5.69 (d, 1H), 4.33-4.20 (m, 2H), 4.12-4.04 (m, 2H), 3.93-3.89 (m, 1H), 2.64-2.54 (m, 1H), 2.29 (s, 3H), 1.11-1.05 (d, 6H). $^{13}\text{C-NMR}$ (75 MHz, $\text{DMSO-}d_6$): δ 176.40, 159.64, 147.65, 145.05, 144.87, 138.82, 128.76, 125.93, 94.65, 90.99, 81.64, 73.73, 69.71, 63.90, 33.60, 21.25, 19.24, 19.19. ESI-MS: 314.18 $[\text{M}+\text{H}]^+$ (Free base M.W.: 313.31).

Example-2: Preparation of 5-isobutrylcytidine tosylate (IV) in two stages :

[0026] Cytidine acetonide sulphate (II) (30.3 g), 4-dimethylaminopyridine (1.94 g, 0.2 equiv.), 1,4-diazabicyclo[5.4.0]undec-7-ene (37.5 g, 3.1 equiv.), were dissolved in acetonitrile (300 mL) and isobutyric anhydride (20.1 g, 1.6 equiv.) was added dropwise and the reaction was completed as described in Example-1 to obtain 29.1 g foamy solid containing 88.17 % (III), 7.31% Impurity-1 and 0.43% an unknown impurity-2 in HPLC analysis.

[0027] The above solid was dissolved in ethyl acetate (150 mL), to this was added a solution of para-toluenesulphonic acid monohydrate (17.23 g, 1.1 equiv.) in ethyl acetate (100 mL). The reaction mixture was stirred at 25 to 30 °C for one hour and the precipitated solids were filtered and washed with ethyl acetate (100 mL), dried under vacuum for 4 hours to obtain 32.8 g (78.54 %) of tosylate salt of III. Purity by HPLC: 97.4 %, IV: 1.98%, Impurity-1: nil. M.R: 160-169°C.

[0028] FT-IR (KBr, cm^{-1}): 3256, 3059, 2987, 2941, 2786, 1731, 1702, 1660, 1543, 1496, 1458, 1409, 1388, 1373, 1352, 1326, 1265, 1252, 1233, 1205, 1191, 1166, 1154, 1124, 1111, 1092, 1072, 1034, 1009, 970, 899, 868, 817, 755, 681, 623, 593, 577, 564. $^1\text{H-NMR}$ (300 MHz, $\text{DMSO-}d_6$): δ 9.51 (s, 1H), 8.42 (s, 1H), 8.00-7.98 (d, 1H), 7.49-7.47 (d, 2H, Ar-H), 7.13-7.10 (d, 2H, Ar-H), 6.08-6.06 (d, 1H), 5.80-5.79 (d, 1H), 5.07-5.04 (dd, 1H), 4.80-4.77 (dd, 1H), 4.36-4.32 (m, 1H), 4.28-4.17 (m, 2H), 2.54-2.47 (m, 1H and DMSO), 2.29 (s, 3H), 1.49 (s, 3H), 1.29 (s, 3H), 1.08-1.05 (d, 6H). $^{13}\text{C-NMR}$ (75 MHz, $\text{DMSO-}d_6$): δ 176.27, 159.93, 147.68, 146.80, 145.46, 138.55, 128.68, 125.93, 113.63, 94.44, 85.48, 84.38, 81.12, 64.21, 33.51, 27.32, 25.54, 21.25, 19.19, 19.10. ESI-MS: 354.12 $[\text{M}+\text{H}]^+$ (Free bas M.W.: 353.37).

[0029] The above solid, tosylate salt of III (32.7g), was suspended in ethyl acetate (300 mL) and heated to 75-80°C and stirred for 20-24 hours. After cooling to room temperature, the solids were filtered and washed with ethyl acetate (100 mL) and acetone (100 mL), dried under vacuum for 3-4 hours to obtain 21.5 g of IV. Purity by HPLC: 98.8%, O-isopropylidene cytidine-5-isobutyrate (III): 0.16%.

Example-3: Preparation of 5-isobutrylcytidine tosylate (IV) :

[0030] The experiment was carried out as described in Example-1, except that the solvent ethyl acetate was replaced with acetonitrile (Yield: 80.6%; Purity by HPLC: 98.6%).

Example-4: Preparation of 2',3'-O-isopropylidene cytidine-5'-isobutyrate (III) :

[0031] Cytidine acetonide sulphate (II) (3g), 4-dimethylaminopyridine (0.19 g, 0.2 equiv.), diisopropylethylamine (DIPEA) (3.15g, 3.1 equiv.), were dissolved in acetonitrile (30 mL) and isobutyric anhydride (3.15 g, 1.6 equiv.) was added during 15 min. The solution was stirred for 16hours. The reaction mixture was concentrated and the residue was

treated with water (100 mL) and dichloromethane (100 mL). After stirring for 5 minutes, the two layers were separated and the organic layer was dried over anhydrous sodium sulphate and concentrated to get 2.6 g (95%) (III). Purity by HPLC: 83.15%; 9.04 % Impurity-1 and 0.31% an unknown impurity-2

Example-5: Preparation of Cytidine-5'-isobutyrate free base (V) :

[0032] Cytidine-5'-isobutyrate tosylate salt (IV) (15 g, 0.031 mol) was suspended in acetone (300 mL) and stirred. To the suspension was added dicyclohexylamine (6.72 g, 0.037 mol) and stirring continued for 2 hours. The precipitated salt of dicyclohexylammonium tosylate was removed by filtration. The solids were washed with acetone (50 mL). The acetone filtrates were pooled and concentrated under reduced pressure. The residue was dissolved in water (100 mL) and washed with dichloromethane (2 x 100 mL). The aqueous solution was concentrated under reduced pressure and the solid residue obtained was azeotroped with acetone to get 8.3 g (85%Y) of foamy solid (V). Purity by HPLC: 99.5%.

Example-6: Preparation of Cytidine-5'-isobutyrate free base (V) :

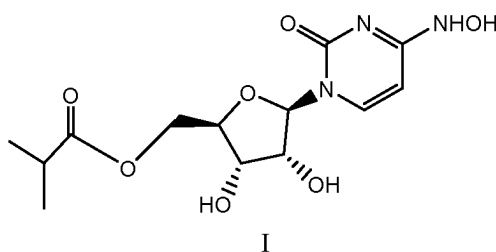
[0033] The experiment was carried out as described in Example-4, except that acetone was replaced with 1,4-Dioxane to obtain 7.8 g (80.6%Y) of V. Purity by HPLC: 98.6 %.

Example-7: Preparation of Molnupiravir (1) :

[0034] Isobutyrylcytidine (V) (5.0g, 0.15 mmol) was suspended in 70% isopropyl alcohol and added hydroxylamine sulphate (8.4g, 0.51 mmol). The reaction mixture was heated to 75-80 °C for 16 hours. The isopropyl alcohol layer was separated and concentrated under reduced pressure. The residue was dissolved in isopropyl alcohol (60 mL) and removed the undissolved matter by filtration and the filtrate was concentrated and the residue was again dissolved in isopropyl alcohol at 50°C. The clear solution was cooled and the solids filtered to obtain 3.7g (71%) (I) as colourless solid, HPLC: 99.29.

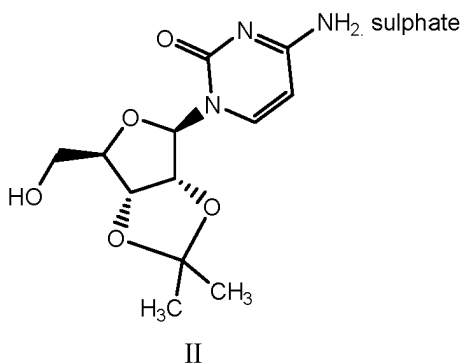
Claims

1. A process for the preparation of Molnupiravir having the formula (I),



comprising:

(a) reacting 2',3'-O-isopropylidene cytidine sulphate having the structure II,



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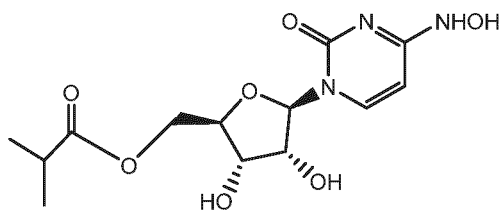


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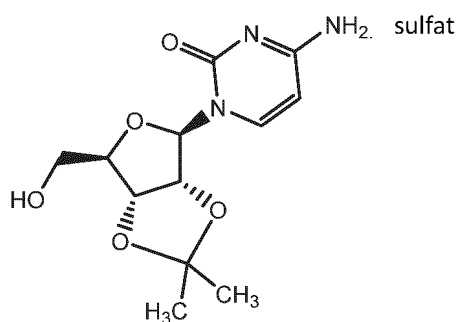
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I

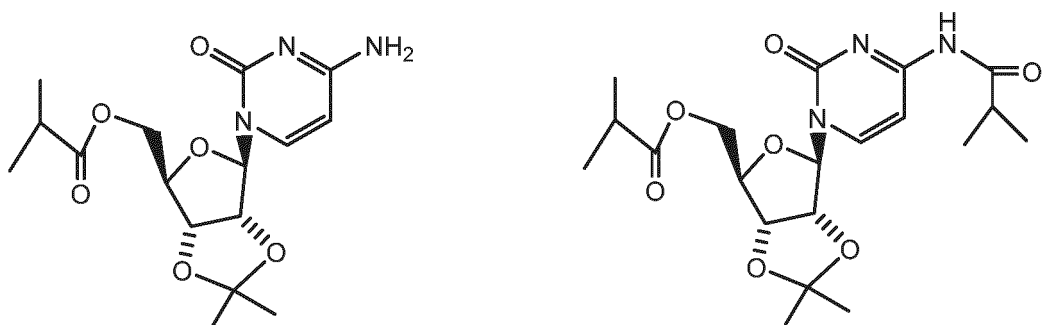
umfassend:

(a) Umsetzen von 2',3'-O-Isopropylidencytidinsulfat mit der Struktur II



II

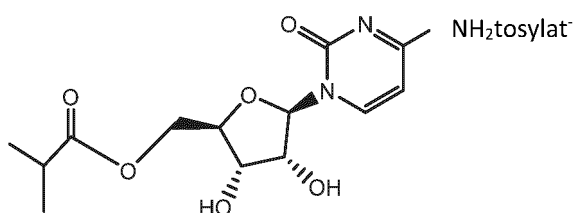
mit Isobuttersäureanhydrid in einem Lösungsmittel in der Gegenwart einer nichtnukleophilen Base, ausgewählt aus der Gruppe, bestehend aus Diisopropylethylamin (DIPEA), N-Methylmorpholin, 1,4-Diazabicyclo-[2.2.2]octan (DABCO), 1.8-Diazabicyclo[5.4.0]undec-7-en (DBU) und Triethylamin, bei Raumtemperatur zum Erhalten von 5-Isobuttersäureester mit der Struktur (III), der N-Isobuttersäureamid als eine Verunreinigung aufweist (Verunreinigung 1),



III

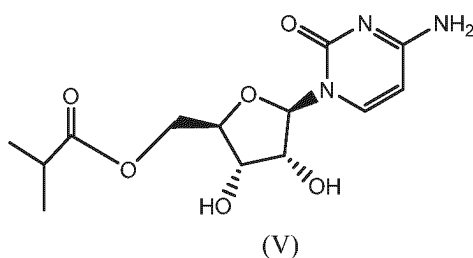
Verunreinigung-1

(b) Umsetzen des Reaktionsprodukts, das im Schritt a erhalten worden ist, mit para-Toluolsulfonsäuremonohydrat zum Erhalten von 5'-Isobutyrylcytidinsäuretosylatsalz (IV), das frei von N-Isobuttersäureamid-Verunreinigung ist, und



(IV)

(c) Umwandeln des 5'-Isobutyrylcytidinsäuresalzes (IV) in dessen freie Base (V)

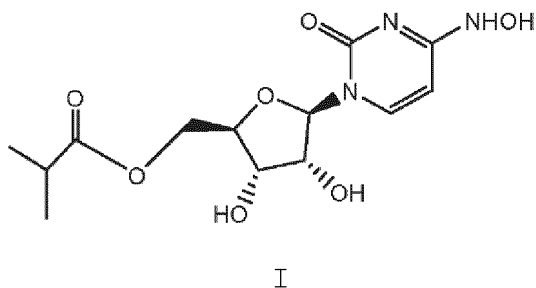


und Umsetzen von (V) mit Hydroxylamin zum Erhalten von Molnupiravir mit der Struktur (I).

2. Verfahren nach Anspruch 1, wobei im Schritt b die Reaktion bei einer Temperatur von 50 °C bis 90 °C durchgeführt wird.
3. Verfahren nach Anspruch 1, wobei im Schritt b das para-Toluolsulfonsäuremonohydrat 8 bis 15 % Hydratisierungswasser aufweist.
4. Verfahren nach Anspruch 1, wobei im Schritt c das 5'-Isobutyrylcytidinsäuresalz (IV) unter Verwendung von Dicyclohexylamin in die freie Base (V) umgewandelt wird.
5. Verfahren nach Anspruch 1, wobei im Schritt c das 5'-Isobutyrylcytidinsäuresalz (IV) unter Verwendung von Aceton oder 1,4-Dioxan als das Lösungsmittel in die freie Base (V) umgewandelt wird.

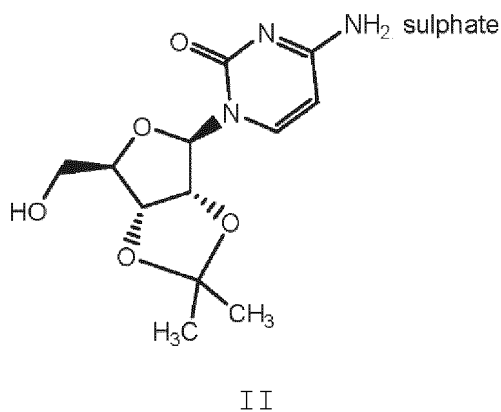
Revendications

1. Procédé de préparation du molnupiravir de formule (I),

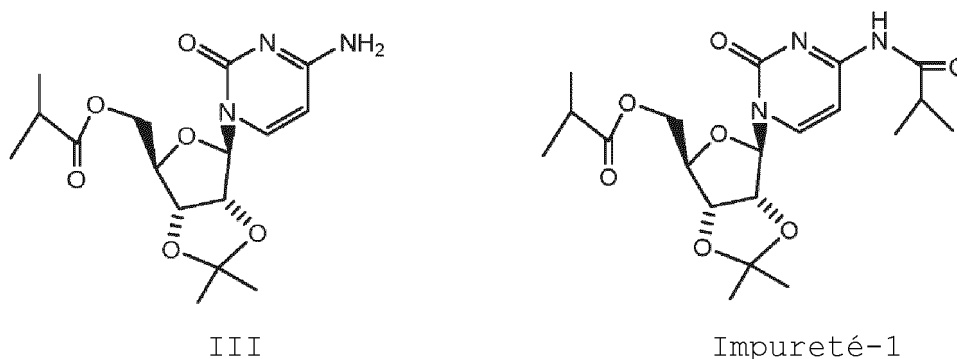


comprenant :

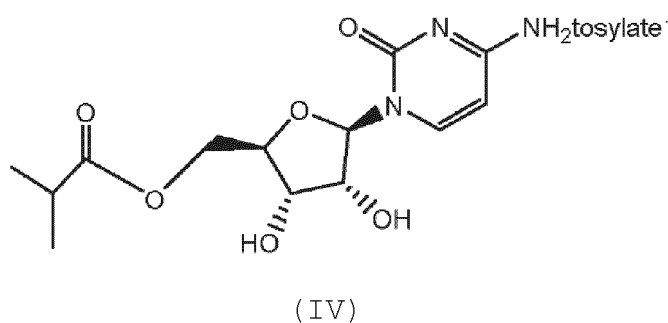
(a) la réaction du sulfate de 2',3'-O-isopropylidène cytidine ayant la structure II,



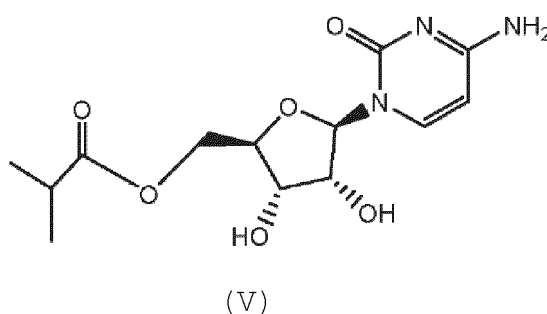
avec de l'anhydride isobutyrique dans un solvant en présence d'une base non nucléophile sélectionnée dans le groupe consistant en la diisopropyléthylamine (DIPEA), le N-méthylmorpholine, le 1,4-diazabicyclo-[2.2.2] octane (DABCO), le 1,8-diazabicyclo[5.4.0]undéc-7-ène (DBU), et la triéthylamine à température ambiante pour obtenir un ester 5-isobutyrique de structure (III), ayant un amide N-isobutyrique en tant qu'impureté (Impureté 1),



(b) la réaction du produit de réaction obtenu dans l'étape-a avec de l'acide para-toluènesulfonique monohydraté pour obtenir le sel de tosylate de l'acide 5'-isobutyryl cytidine (IV), qui est exempt de l'impureté amide N-isobutyrique, et



(c) la conversion du sel d'acide 5'-isobutyryl cytidine (IV) en sa base libre (V)



et la réaction de (V) avec de l'hydroxylamine pour obtenir le molnupiravir de structure (I).

2. Procédé selon la revendication 1, dans lequel dans l'étape b, la réaction est menée à une température de 50 °C à 90 °C.
3. Procédé selon la revendication 1, dans lequel dans l'étape b, l'acide para-toluène sulfonique monohydraté présente 8 à 15 % d'eau d'hydratation.
4. Procédé selon la revendication 1, dans lequel dans l'étape c, le sel d'acide 5'-isobutyryl cytidine (IV) est converti en base libre (V) en utilisant la dicyclohexylamine.

5. Procédé selon la revendication 1, dans lequel dans l'étape c, le sel d'acide 5'-isobutyryl cytidine (IV) est converti en base libre (V) en utilisant de l'acétone ou du 1,4-dioxane comme solvant.

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REFERENCES CITED IN THE DESCRIPTION

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Patent documents cited in the description

- WO 2019113462 A [0004] [0008]
- WO 2019173602 A [0004] [0008]
- WO 2016106050 A1 [0008]

Non-patent literature cited in the description

- *Chem. Commun.*, 2020, vol. 56, 13363-13364 [0005]
- *Synlett.*, 2021, vol. 32 (3), 326-328 [0007]
- *Synlett*, 2021, vol. 32 (3), 326-328 [0017]
- *Biochemical & Biophysical research Communications*, 1965, vol. 18 (4), 617-622 [0020]